① $\int_{-1}^{1} e^{1/x}\frac{dx}{x^3}$ – особеност в $x=0$ – ф-ята е неограничена. Тогава $\int_{-1}^{1}=\int_{-1}^{0}+\int_{0}^{1}$

$\int_{-1}^{0} e^{1/x}\frac{dx}{x^3}=\left(\frac{1}{x}=y\right)=\int_{-1}^{-\infty} e^{y}y^{3}d\frac{1}{y}=\int_{-\infty}^{-1} e^{y}y\,dy=(y=-t)=\int_{1}^{\infty} e^{-t}t\,dt=\int_{1}^{\infty}\frac{dt}{e^{t}t^{-1}}$ – сходящ – сравнете с ⊛ $\int_{a}^{\infty}\frac{dx}{a^{x}x^{\alpha}\ln^{\beta}x}$, $a>1\to$ сх.

$\int_{0}^{1}=\left(\frac{1}{x}=y\right)=\int_{1}^{\infty} e^{y}y\,dy$ – разходящ – сравнете с ⊛. Окончателно $\int_{-1}^{1}$ – разходящ.

② $\int_{-1}^{1}\frac{dx}{\ln(1+x)}$ – особеност в 0-та. В т. $x=-1\to\ln(1+x)\xrightarrow[x\to-1]{}-\infty$, т.е. $\frac{1}{\ln(1+x)}\xrightarrow[x\to-1]{}0$ и $x=-1$ не е особена.

$\int_{-1}^{1}=\int_{-1}^{0}+\int_{0}^{1}$. В т. $x=0\to\ln(1+x)\underset{x\to0}{\sim}x\Rightarrow\int_{-1}^{0}\frac{dx}{\ln(1+x)}\sim\int_{-1}^{0}\frac{dx}{x}$ – разходящ $\Rightarrow\int_{-1}^{1}$ е разходящ.

За да е сходящ $\int_{-1}^{1}$ трябва и двата интеграла $\int_{-1}^{0}$ и $\int_{0}^{1}$ да са сходящи.

③ $\int_{0}^{1}\frac{\ln x\,dx}{\sqrt{x(1-x)^3}}$ – особености в $x=0$ и $x=1$. Следователно $\int_{0}^{1}=\int_{0}^{1/2}+\int_{1/2}^{1}$

$\int_{0}^{1/2}\frac{\ln x\,dx}{\sqrt{x(1-x)^3}}\sim\int_{0}^{1/2}\frac{\ln x\,dx}{\sqrt{x}}=\int_{0}^{1/2}\frac{dx}{x^{1/2}\ln^{-1}x}$ – сходящ – сравнете с $\int_{0}^{a}\frac{dx}{x^{\alpha}\ln^{\beta}x}$, който е сходящ при $\alpha<1$.

$\int_{1/2}^{1}\frac{\ln x\,dx}{\sqrt{x(1-x)^3}}\sim\int_{1/2}^{1}\frac{\ln x\,dx}{(1-x)^{3/2}}=(1-x=y)=-\int_{1/2}^{0}\frac{\ln(1-y)}{y^{3/2}}dy=\int_{0}^{1/2}\frac{\ln(1-y)}{y^{3/2}}dy.$

Но $\ln(1-y)\underset{y\to0}{\sim}y\Rightarrow\int_{0}^{1/2}\sim\int_{0}^{1/2}\frac{y}{y^{3/2}}dy=\int_{0}^{1/2}\frac{dy}{y^{1/2}}$ – сходящ. Окончателно $\int_{0}^{1}$ – сходящ.

④ $\int_{0}^{1}\frac{\arcsin(x^2+x^3)}{x\ln^{2}(1+x)}dx$ – особеност в $x=0$

$\int_{0}^{1}\sim\int_{0}^{1}\frac{\arcsin(x^2)}{x.x^2}dx\sim\int_{0}^{1}\frac{x^2}{x^3}dx=\int_{0}^{1}\frac{dx}{x}\to$ разходящ

Обяснете: $\ln(1+x)\underset{x\to0}{\sim}x$, $\arcsin x\underset{x\to0}{\sim}x$, $x^2+x^3\underset{x\to0}{\sim}x^2$

⑤ $\int_{0}^{1}e^{\alpha/x}(\cos x)^{1/x^3}dx$ – особеност в $x=0$.

$(\cos x)^{1/x^3}=e^{\ln(\cos x)^{1/x^3}}=e^{1/x^3\ln\cos x}$

$\cos x\underset{x\to0}{\sim}1-\frac{x^2}{2}\Rightarrow\ln\cos x\underset{x\to0}{\sim}\ln\left(1-\frac{x^2}{2}\right)\sim-\frac{x^2}{2}\Rightarrow e^{1/x^3\ln\cos x}\sim e^{-\frac{1}{2x}}$. Тогава

$\int_{0}^{1}\sim\int_{0}^{1}e^{\alpha/x}.e^{-1/2x}dx=\int_{0}^{1}e^{(\alpha-1/2)1/x}dx=\left(\frac{1}{x}=y\right)=\int_{\infty}^{1}e^{(\alpha-1/2)y}d\frac{1}{y}=\int_{1}^{\infty}\frac{e^{(\alpha-1/2)y}}{y^2}dy=\int_{1}^{\infty}\frac{dy}{e^{(1/2-\alpha)y}y^2}$

① $\alpha=\frac{1}{2}\to\int_{1}^{\infty}\frac{dy}{y^2}\to$ сходящ

② $\alpha<\frac{1}{2}\to\int_{1}^{\infty}\frac{dy}{e^{(1/2-\alpha)y}y^2}<\int_{1}^{\infty}\frac{dy}{y^2}\Rightarrow$ сходящ

③ $\alpha>\frac{1}{2}\to$ тогава $e^{1/2-\alpha}<e^{0}=1\Rightarrow$ разходящ – следва от сравняване с $\int^{\infty}\frac{dx}{a^{x}x^{\alpha}\ln^{\beta}x}$, където, в случая, $a<1$.

⑥ $\int_{0}^{1}\frac{\sqrt{e^2+x^2}-e^{\cos x}}{x^{\alpha}}dx$ – особеност единствено при $x=0$.

$$\begin{cases}\sqrt{e^2+x^2}=e\sqrt{1+\left(\frac{x}{e}\right)^2}=e\left(1+\left(\frac{x}{e}\right)^2\right)^{1/2}\sim e\left(1+\frac{1}{2}\left(\frac{x}{e}\right)^2+\dots\right)=e+\frac{x^2}{2e}+\dots\\ e^{\cos x}=e^{1-x^2/2+\dots}=e.e^{-x^2/2+\dots}=e\left(1-\frac{x^2}{2}+\dots\right)=e-\frac{ex^2}{2}+\dots\end{cases}$$

$\sqrt{e^2+x^2}-e^{\cos x}\sim\left(e+\frac{x^2}{2e}+\dots\right)-\left(e-\frac{ex^2}{2}+\dots\right)=\left(\frac{1}{2e}+\frac{e}{2}\right)x^2+\dots\sim x^2$

$\Rightarrow\int_{0}^{1}\sim\int_{0}^{1}\frac{x^2}{x^{\alpha}}dx=\int_{0}^{1}\frac{dx}{x^{\alpha-2}}\Rightarrow$ сходящ при $\alpha-2<1$, т.е. $\alpha<3$.

⑦ $\int_{0}^{\pi/2}\frac{\cos^2(2x)-e^{-4x^2}}{x^{\alpha}\operatorname{tg}x}dx$ – особености в 0-та и $\pi/2$, понеже $\lim_{x\to\pi/2-}\operatorname{tg}x=\infty$.

$\int_{0}^{\pi/2}=\int_{0}^{1}+\int_{1}^{\pi/2}$, $\int_{0}^{1}\to$ $\cos^2(2x)=\left(1-\frac{(2x)^2}{2}+\frac{1}{4!}(2x)^4+\dots\right)^2=1-4x^2+4x^4+\frac{2^4}{4!}x^4+O(x^6)$

$e^{-4x^2}=1-4x^2+\frac{1}{2}(4x^2)^2+O(x^6)$, $\operatorname{tg}x\underset{x\to0}{\sim}x$

$\Rightarrow\cos^2(2x)-e^{-4x^2}=\left(4+\frac{2^4}{4!}-8\right)x^4+O(x^6)\sim x^4\Rightarrow$

$\int_{0}^{1}\sim\int_{0}^{1}\frac{x^4}{x^{\alpha+1}}dx=\int_{0}^{1}\frac{dx}{x^{\alpha-3}}\Rightarrow$ сходящ при $\alpha-3<1$, т.е. $\alpha<4$.

$\int_{1}^{\pi/2}\sim\int_{1}^{\pi/2}\frac{\cos^2(2x)-e^{-4x^2}}{\operatorname{tg}x}dx$, понеже $x^{\alpha}\underset{x\to\pi/2}{\sim}const$

$\lim_{x\to\pi/2-}\left(\cos^2(2x)-e^{-4x^2}\right)=1-e^{-\pi^2}\neq0\Rightarrow\cos^2(2x)-e^{-4x^2}\underset{x\to\pi/2}{\sim}const$

$\Rightarrow\int_{1}^{\pi/2}\sim\int_{1}^{\pi/2}\frac{dx}{\operatorname{tg}x}$ – сходящ, понеже $\lim_{x\to\pi/2-}\frac{1}{\operatorname{tg}x}=0$, т.е. ф-ята е ограничена или интегралът е собствен.

Окончателно $\int_{0}^{\pi/2}$ е сходящ при $\alpha<4$.

⑧ $\int_{0}^{\pi/2}\frac{e^{\alpha\cos x}-\sqrt{1+2\cos x}}{\sqrt{\cos^5 x}}dx$ – особеност единствено при $x=\pi/2$

$\int_{0}^{\pi/2}=\left(x=\frac{\pi}{2}-y\right)=-\int_{\pi/2}^{0}\frac{e^{\alpha\cos(\pi/2-y)}-\sqrt{1+2\cos(\pi/2-y)}}{\sqrt{\cos(\pi/2-y)^5}}dy=\int_{0}^{\pi/2}\frac{e^{\alpha\sin y}-\sqrt{1+2\sin y}}{(\sin y)^{5/2}}dy$

Последният интеграл има единствена особеност в 0-та.

$$\left.\begin{cases}e^{\alpha\sin y}=1+\alpha\sin y+\frac{1}{2}(\alpha\sin y)^2+\dots\\ (1+2\sin y)^{1/2}=1+\sin y+\binom{1/2}{2}(2\sin y)^2+\dots\end{cases}\right\}\Rightarrow e^{\alpha\sin y}-\sqrt{1+2\sin y}=(\alpha-1)\sin y+\left(\frac{\alpha^2}{2}+\frac{1}{2}\right)\sin^2 y+\dots$$

① $\alpha\neq1\to\int_{0}^{\pi/2}\sim\int_{0}^{\pi/2}\frac{\sin y}{(\sin y)^{5/2}}dy=\int_{0}^{\pi/2}\frac{dy}{(\sin y)^{3/2}}\sim\int_{0}^{\pi/2}\frac{dy}{y^{3/2}}\to$ разходящ

② $\alpha=1\to\int_{0}^{\pi/2}\sim\int_{0}^{\pi/2}\frac{\sin^2 y}{(\sin y)^{5/2}}dy=\int_{0}^{\pi/2}\frac{dy}{\sqrt{\sin y}}\sim\int_{0}^{\pi/2}\frac{dy}{\sqrt{y}}\to$ сходящ

⑨ $\int_{0}^{1}\frac{\ln(e^{x^{\alpha}}+x)-x}{\operatorname{tg}x}dx$ – особеност в $x=0$. $\operatorname{tg}x\underset{x\to0}{\sim}x$

1) $\alpha<0\to x^{\alpha}\xrightarrow[x\to0]{}\infty\Rightarrow e^{x^{\alpha}}\xrightarrow[x\to0]{}\infty\Rightarrow e^{x^{\alpha}}+x\sim e^{x^{\alpha}}\Rightarrow\ln(e^{x^{\alpha}}+x)\sim\ln e^{x^{\alpha}}=x^{\alpha}\Rightarrow$

$\int_{0}^{1}\sim\int_{0}^{1}\frac{x^{\alpha}-x}{x}dx\sim\int_{0}^{1}\frac{x^{\alpha}}{x}dx=\int_{0}^{1}\frac{dx}{x^{1-\alpha}}$ и от $1-\alpha>1\Rightarrow$ разходящ – сравнете с $\int_{0}^{a}\frac{dx}{x^{\alpha}}$, сх. при $\alpha<1$.

2) $\alpha=0\to\int_{0}^{1}\frac{\ln(e+x)-x}{x}dx\sim\int_{0}^{1}\frac{dx}{x}$ – разходящ

3) $\alpha>0\to e^{x^{\alpha}}=1+x^{\alpha}+\dots\Rightarrow\ln(e^{x^{\alpha}}+x)=\ln(1+x^{\alpha}+x+\dots)$

3.1) $\alpha<1\to\ln(1+x^{\alpha}+x+\dots)\sim x^{\alpha}\Rightarrow\ln(e^{x^{\alpha}}+x)-x\sim x^{\alpha}-x\sim x^{\alpha}\Rightarrow$

$\int_{0}^{1}\sim\int_{0}^{1}\frac{x^{\alpha}}{x}dx=\int_{0}^{1}\frac{dx}{x^{1-\alpha}}\to$ сходящ, понеже при $0<\alpha<1\to1-\alpha<1$

3.2) $\alpha=1\to\ln(e^{x^{\alpha}}+x)=\ln(e^{x}+x)=\ln\left(1+x+\frac{x^2}{2}+\dots+x\right)=\ln(1+2x+\dots)\sim2x$

$\Rightarrow\int_{0}^{1}\sim\int_{0}^{1}\frac{2x-x}{x}dx=1\to$ сходящ

3.3) $\alpha>1\to\ln(e^{x^{\alpha}}+x)=\ln\left(1+x^{\alpha}+\frac{1}{2}x^{2\alpha}+\dots+x\right)\sim x+x^{\alpha}\Rightarrow$

$\int_{0}^{1}\sim\int_{0}^{1}\frac{x+x^{\alpha}-x}{x}dx=\int_{0}^{1}\frac{dx}{x^{1-\alpha}}$ – сходящ

Окончателно $\int_{0}^{1}$ е сходящ при $\alpha>0$.